

Cervical Cytology-Histology Correlation Quality Assurance Report

Automated pipeline per ASC Birdsong Guideline 2017 | CAP CYP.06600

Total Cases Analyzed: 130

Executive Summary

A total of 130 cases were analyzed.

CAP ALERT: Major discordance rate of 26.1% exceeds the CAP benchmark of less than 10%. Corrective action documentation required per CYP.06600.

PV+ ALERT: HSIL positive predictive value of 42.3% falls below the CAP target of 60%.

Section 1 ? Concordance Summary

Classification of all cytology-histology pairs per ASC Birdsong concordance definitions.

Concordance_Class	Count	Percent
minor_discordant	65	50.0
major_discordant	34	26.15
concordant	31	23.85

Agreement Within One Grade (Birdsong Section 5)

Percentage of pairs with exact agreement or minor disagreement only. Per Birdsong, this metric is of educational interest.

Metric	Count	Percentage
Agreement within one grade	96	73.8%
Exact agreement only	31	23.9%
Major discordance (outside one grade)	34	26.1%

Section 2 ? Discrepancy Bucket Summary

CAP-style discrepancy classification per Birdsong Figure 1 and Figure 3.

Bucket	Count	Percent
MinVar	7	5.38
MajUnd	14	10.77
MinUnd	43	33.08
Agree	31	23.85
MinOver	15	11.54
MajOver	20	15.38

Section 3 ? HSIL-Focused Metrics (Birdsong Section 7)

The five statistical calculations required by Birdsong Section 7 and the ASC Clinical Practice Committee.

Metric	Value
HSIL Pap tests with histologic HSIL (CIN2/CIN3)	42.31
HSIL Pap tests with minor discrepancies	23.08

Metric	Value
HSIL Pap tests with major false-positive discrepancies	34.62
Number of major false-negative / cytology undercall discrepancies	14.0
PV+ for HSIL Pap tests (%)	42.31

PV+ Interpretation (Birdsong Section 7a)

LOW ? Below CAP target of 60%. Review HSIL overcall rate.

Extended PV+ ? High-Grade Cytology Group (HSIL + ASC-H + AGC-NEO)

Per Birdsong Figure 1, HSIL, ASC-H, and AGC-NEO are grouped together for concordance assessment. The extended PV+ reflects combined performance.

Metric	Value
Total high-grade Pap tests (HSIL + ASC-H + AGC-NEO)	61
High-grade Pap tests with HSIL+ histology	31
Extended PV+	50.8%

Section 4 ? Intergrade Follow-Up Rates (Birdsong Section 8)

Per Birdsong Section 8: review of HSIL histology rates following ASC-US, ASC-H, and LSIL cytologic interpretations to identify systematic undercall patterns.

Cytology	Total Cases	HSIL+ Follow-up	HSIL+ Rate (%)
ASC-US	11	9	81.8
ASC-H	11	3	27.3
LSIL	17	11	64.7
AGC	14	10	71.4
AGC-NEO	9	5	55.6
AGC-EMC	4	4	100.0

Elevated HSIL+ follow-up rates in ASC-US or LSIL categories may indicate systematic cytologic undercalling requiring targeted slide review per Birdsong Section 8.

Section 5 ? Key Quality Signals

Quality Signal	Value	CAP Threshold	Status
Major discordance rate	26.1%	< 10%	EXCEEDS
Minor discordance rate	50.0%	?	?
Undercall proportion	43.9%	< 15%	EXCEEDS
Overcall proportion	26.9%	?	?
HSIL PV+	42.3%	> 60%	LOW
Extended PV+ (HSIL+ASC-H+AGC-NEO)	50.8%	> 60%	?
Agreement within one grade	73.8%	?	?
Major false negative count	14	Minimize	REVIEW

Section 6 ? Visualizations

Figure 1 ? Concordance Distribution

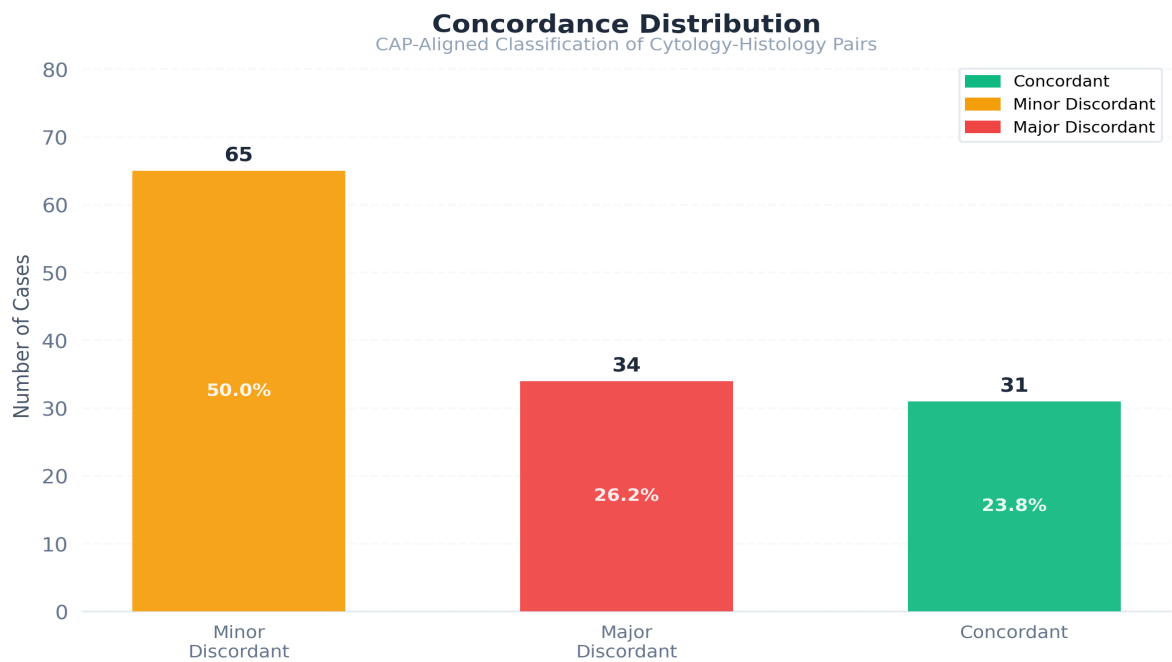


Figure 2 ? Cytology-Histology Correlation Buckets

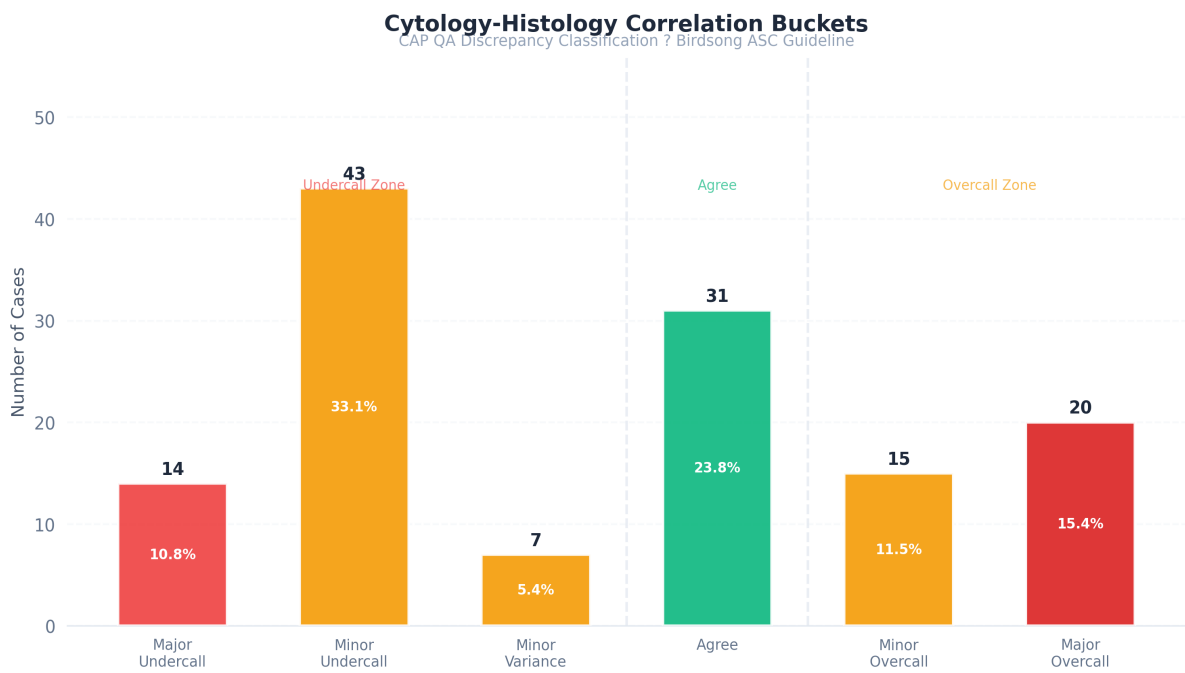


Figure 3 ? HSIL Correlation Metrics

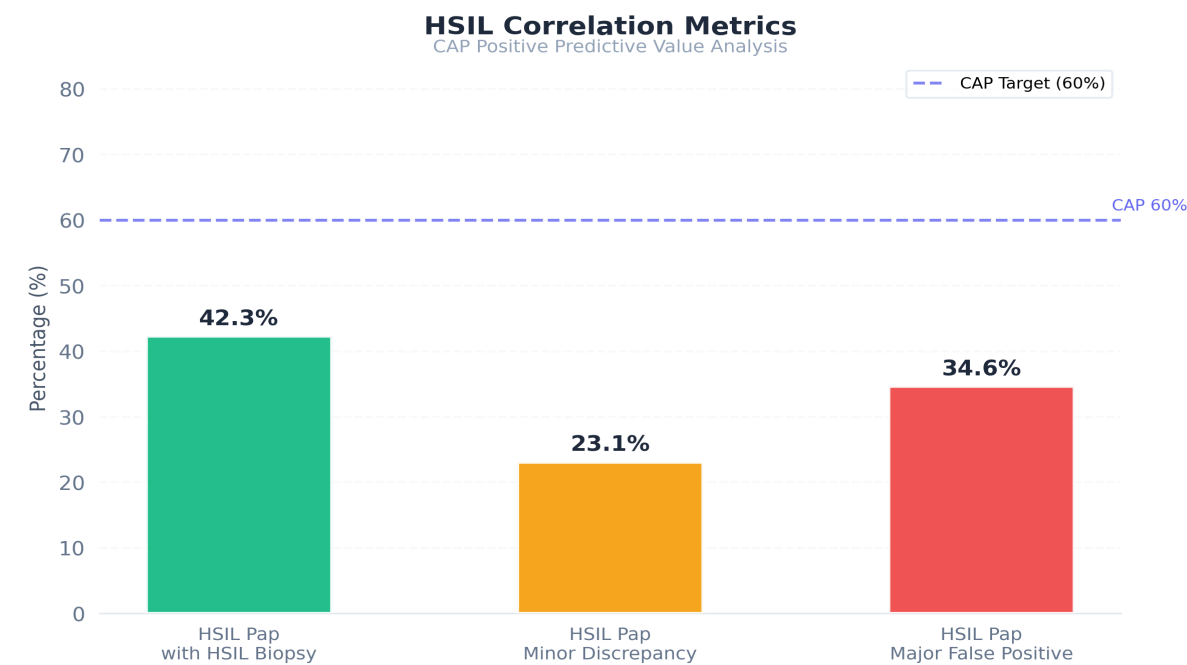


Figure 4 ? Cytology vs Histology Confusion Matrix

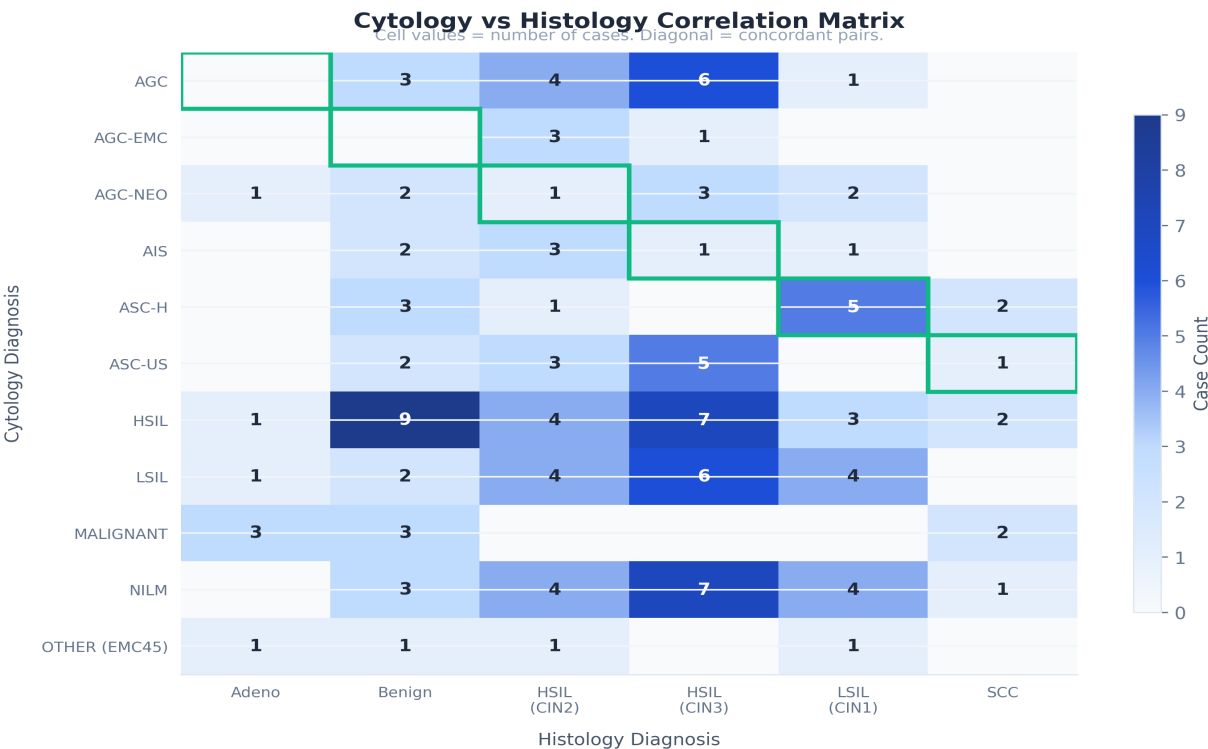
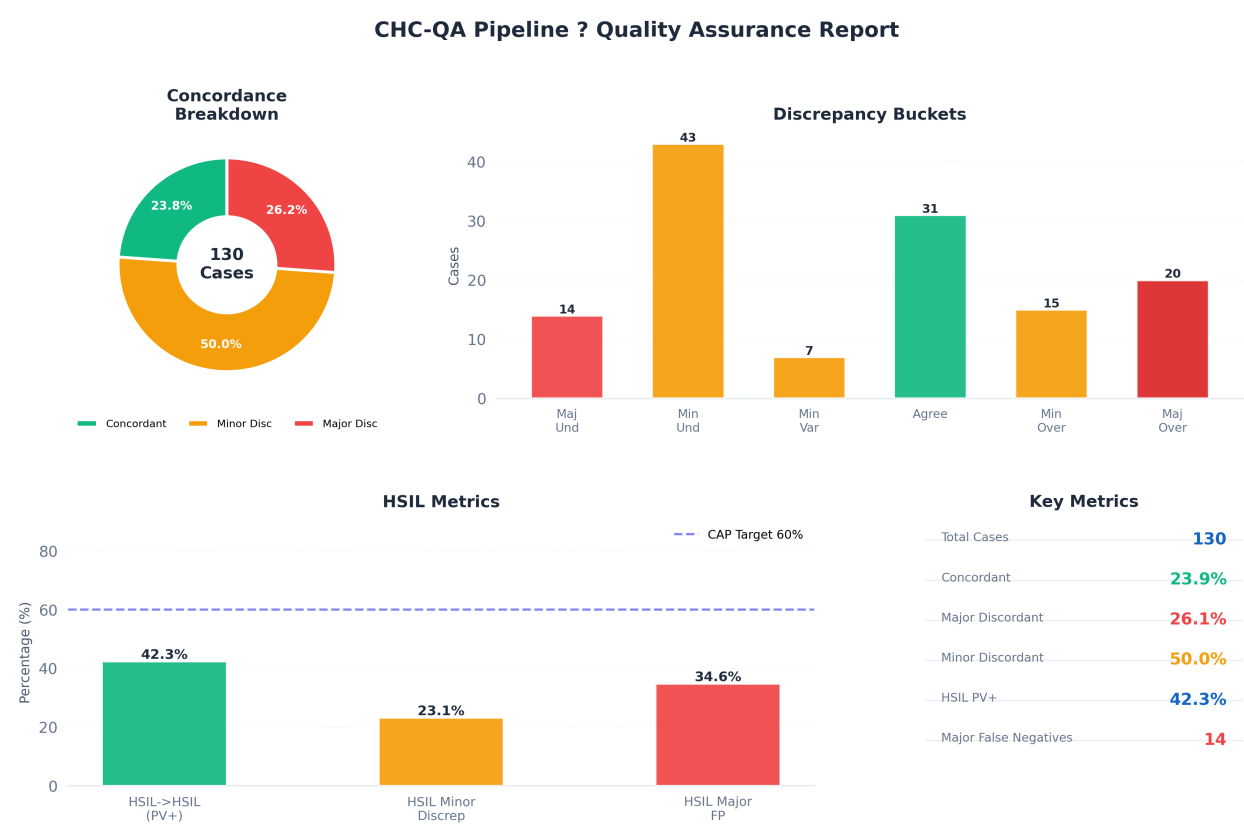


Figure 5 ? Summary Dashboard



Section 8 ? Limitations

This report is generated by an automated deterministic cytology-histology correlation pipeline. Classification rules are derived from the ASC Clinical Practice Committee Birdsong Guideline (2017) and applied deterministically to all pairs. The following limitations apply:

1. Verification bias: CHC datasets are subject to verification bias due to selective histologic follow-up. Results should not be interpreted as measures of screening sensitivity or specificity.
2. Observed discrepancies should be interpreted within the context of laboratory QA review rather than as direct indicators of diagnostic error.
3. LLM normalization: Free-text diagnosis normalization was performed by a locally-hosted large language model. Normalized terms were validated against a controlled vocabulary. Residual normalization errors may affect a small number of cases.
4. Pairing logic: Cases were paired by patient identifier and date window. Complex scenarios including multiple biopsies, multiple Pap tests, and delayed follow-up may result in suboptimal pairing in a minority of cases.
5. Root cause classification: This pipeline classifies discordance type and direction but does not perform root cause analysis (sampling error, screening error, interpretive error). Manual review of major discordant cases is required per Birdsong Section 8.

Reference: Birdsong GG, Walker JW. Gynecologic Cytology-Histology Correlation Guideline. ASC Bulletin. 2017;LIV(2):VIII-XIII.